

SNOWFLAKE

End-to-End Data Pipeline Project

A comprehensive summary of the project architecture, components, data flow, and key learnings

Project Overview

This project implements a complete end-to-end data pipeline entirely within Snowflake. Raw data is ingested from source systems, progressively cleaned and transformed through multiple layers, and finally made available for business analytics and reporting. The pipeline supports both batch and incremental data loading, is fully automated using native Snowflake features, and follows the Medallion Architecture (Bronze, Silver, Gold) to ensure data quality and reliability at every stage.

Pipeline Architecture — Layer by Layer

Layer	Purpose	Snowflake Feature / Tool
1. Ingestion (Bronze Layer)	Load raw source data into Snowflake without any transformation	Snowpipe, COPY INTO, Internal / External Stage
2. Staging (Silver Layer)	Clean the data — remove duplicates, fix data types, handle nulls	Streams, Tasks, SQL Transformations, dbt
3. Curated (Gold Layer)	Build business-ready aggregated tables by joining multiple sources	Dynamic Tables, Materialized Views
4. Analytics Layer	Expose final views to BI tools and business users securely	Secure Views, Data Shares, Power BI / Tableau

Key Components — What Was Built and Why

1. Snowpipe — Automated Data Ingestion

- Configured Snowpipe to automatically load files into Snowflake as soon as they land in cloud storage (AWS S3 / Azure Blob).
- Set up SQS event notifications to trigger the pipe in near real-time without any manual intervention.
- Defined the pipe using a COPY INTO command — fully serverless and cost-efficient.

2. Streams and Tasks — Incremental Processing (CDC)

- Used Snowflake Streams to track every INSERT, UPDATE, and DELETE on source tables — a pattern known as Change Data Capture (CDC).
- Scheduled Tasks to run at defined intervals, processing only the changed rows rather than reloading the entire table.
- This approach significantly reduced compute costs and improved pipeline performance.

3. dbt (Data Build Tool) — SQL Transformations

- Organized all transformation logic into modular dbt models — each model is a separate, version-controlled SQL file.
- Used dbt's lineage graph to clearly visualize how data flows from raw sources to final business tables.
- Wrote data tests to automatically validate NULL values, duplicates, and referential integrity after every run.

4. Dynamic Tables — Auto-Refreshing Gold Layer

- Used Snowflake Dynamic Tables in the Gold layer to define transformation logic declaratively.
- Snowflake automatically refreshes these tables based on a configured target lag — no manual scheduling needed.
- Eliminated complex MERGE statements and reduced pipeline maintenance overhead.

5. Role-Based Access Control (RBAC) and Data Security

- Created purpose-specific roles: DATA_ENGINEER, DATA_ANALYST, and REPORTER, each with appropriate permissions.
- Applied Column-Level Security to mask sensitive fields such as PII from unauthorized roles.
- Implemented Row Access Policies so that different users see only the data they are authorized to access from the same table.

6. Monitoring and Alerting

- Monitored pipeline health using QUERY_HISTORY and TASK_HISTORY views in Snowflake.
- Configured Snowflake Alerts to send email notifications whenever a scheduled task fails.
- Set up Resource Monitors to track and control credit consumption, preventing unexpected cost overruns.

Data Flow — Step by Step

Step	Action	Output
Step 1	Source systems generate files (CSV / JSON / Parquet) and upload them to AWS S3 or Azure Blob Storage	Raw files in cloud storage
Step 2	Snowpipe detects new files via SQS notification and executes COPY INTO to load data	Bronze layer — raw tables
Step 3	Snowflake Stream captures new rows; Scheduled Task runs SQL to clean and validate	Silver layer — cleaned tables
Step 4	dbt models join and aggregate Silver tables to produce business metrics	Gold layer — curated tables
Step 5	Dynamic Tables and Secure Views are created on top of the Gold layer	Analytics-ready layer
Step 6	Power BI or Tableau connects via Snowflake connector to render dashboards	Reports and dashboards

Technologies Used

Category	Technology / Tool
Cloud Storage	AWS S3 / Azure Blob Storage
Data Ingestion	Snowpipe, COPY INTO, Internal Stage, External Stage
Incremental Processing	Snowflake Streams, Snowflake Tasks (CDC)
Transformations	dbt Core / dbt Cloud, SQL
File Formats	CSV, JSON, Parquet
Gold Layer	Dynamic Tables, Materialized Views
Access Control	RBAC, Row-Level Security, Column-Level Security
BI and Reporting	Power BI, Tableau, Snowsight
Monitoring	Snowflake Alerts, Resource Monitors, Query History
Version Control	Git with dbt project structure

Key Learnings

1. Gained a strong understanding of the Medallion Architecture (Bronze, Silver, Gold) and why separating data into layers improves quality, maintainability, and performance.
2. Built a fully automated incremental pipeline using the combination of Snowpipe, Streams, and Tasks — without any external orchestration tool.
3. Elevated SQL transformation quality using dbt — implementing testing, documentation, and data lineage in a production-grade manner.
4. Implemented comprehensive data security using Role-Based Access Control, Column-Level Security, and Row Access Policies.
5. Learned cost optimization strategies including correct Virtual Warehouse sizing, auto-suspend configuration, and proper use of clustering keys.
6. Solved real-world engineering challenges such as handling late-arriving data and designing idempotent pipelines that can be safely re-run.